ST5230 Assignment 1

Language Modeling, Embeddings, and Downstream Classification

Overview

This report compares four families of language models -- n-gram, RNN, LSTM, and Transformer -- trained on the same English news corpus under controlled settings. We then investigate how the choice of word embedding affects each neural model, and finally evaluate whether better language models produce better features for a downstream classification task.

All experiments use the AG News dataset, a balanced 4-class topic classification corpus (World, Sports, Business, Sci/Tech) with 96,000 training documents (~4.7M tokens) and 7,600 test documents. We tokenize by whitespace with lowercasing and punctuation separation, retaining the top 20,000 tokens as vocabulary. Sequences are truncated to 128 tokens. For language modeling, we split the training data 80/10/10 into train/val/test; for downstream classification, we use the original HuggingFace test split (1,900 per class).

To ensure fair comparison, all three neural language models share the same dimensionality (embed_dim = hidden_dim = 100), depth (2 layers), regularization (dropout = 0.1), and training regime (Adam, lr = 0.001, gradient clipping at 5.0, batch size 64, 20 epochs). This keeps their total parameter counts within 4.06M to 4.28M, so performance differences reflect architectural choices rather than model capacity.

Part I: Language Model Training and Comparison

We train five language models on the same data: a bigram and trigram (count-based with Laplace smoothing, k = 1), a 2-layer vanilla RNN with tanh activation, a 2-layer LSTM, and a 2-layer decoder-only Transformer (GPT-style, 4 attention heads, d_ff = 400, learned positional embeddings with causal masking).

Model	Architecture	Parameters
Bigram	Count-based, Laplace smoothing	N/A
Trigram	Count-based, Laplace smoothing	N/A
RNN	2-layer vanilla RNN, tanh	4,060,400
LSTM	2-layer LSTM	4,181,600
Transformer	2-layer decoder-only, 4 heads, d_ff=400	4,275,600

Table 1. Model architectures and parameter counts.

Results

Model	Val PPL	Test PPL	Train Time	Inf. (ms/batch)
Bigram	860.44	849.76	5.3s	4.73
Trigram	3791.59	3735.70	10.0s	6.03
RNN	97.46	96.26	1878.5s (~94s/ep)	0.90
LSTM	78.75	77.82	1997.8s (~100s/ep)	0.94
Transformer	54.07	53.59	2101.9s (~105s/ep)	2.26

Table 2. Perplexity, training time, and inference speed.

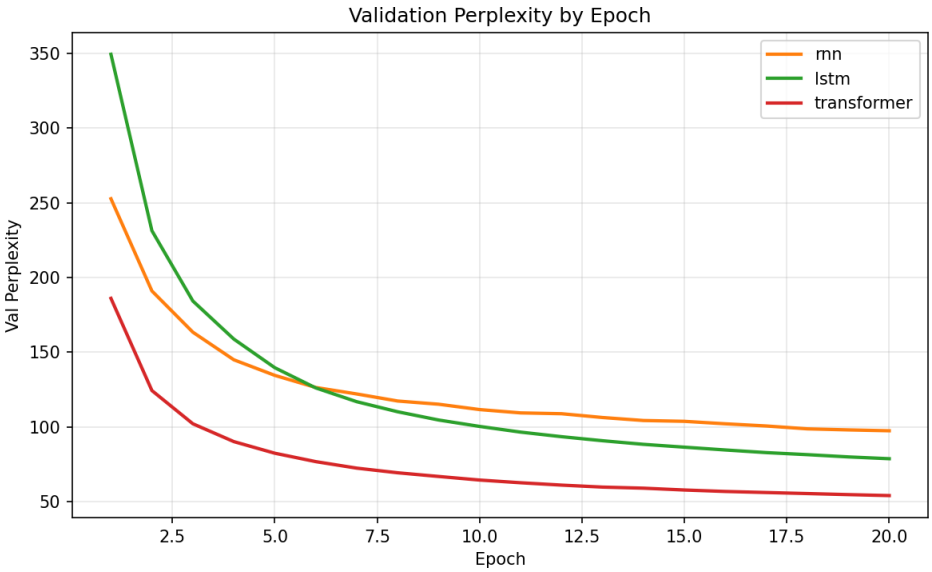

Figure 1. Validation perplexity by epoch. The Transformer converges fastest and reaches the lowest PPL.

Figure 2. Train vs validation loss. The small gap confirms no significant overfitting over 20 epochs.

Analysis

The n-gram models serve as a useful baseline but have far higher perplexity. The trigram (PPL 3736) performs worse than the bigram (PPL 850) because Laplace smoothing distributes mass uniformly across unseen events -- and with 1.0M distinct trigram histories (vs 60K bigram), sparsity is far more severe.

Among neural models, the hierarchy is clear: Transformer (PPL 53.6) > LSTM (PPL 77.8) > RNN (PPL 96.3). The LSTM's 19% improvement over the RNN comes from its gating mechanism, which provides a gradient highway through the cell state. The Transformer extends this further -- self-attention creates direct connections between any two positions, avoiding the sequential bottleneck. Its PPL is 31% lower than LSTM's.

This quality comes at a cost: the Transformer is the slowest model in training (105s/epoch vs 100s for LSTM, 94s for RNN) and inference (2.26 ms/batch vs ~0.9 ms), reflecting O(n^2) attention overhead at this small scale.

In generated text, n-gram samples lose topic consistency within a few words. The RNN drifts after ~15 tokens, the LSTM holds slightly longer, and the Transformer produces the most topically coherent passages.

Part II: Embedding Variants and Ablation

To understand how the embedding layer affects language modeling, we replace the default trainable embeddings with two frozen alternatives and retrain each neural model. This gives a 3 x 3 grid of 9 experiments:

- Scratch: standard trainable embeddings (nn.Embedding, 100-dim), randomly initialized.
- W2V frozen: Word2Vec (Skip-gram, dim=100, window=5) trained on the AG News training set only via Gensim, then frozen. Coverage: 99.98% (19,996/20,000).
- GloVe pretrained frozen: glove-wiki-gigaword-100, a public embedding from Wikipedia + Gigaword. Coverage: 98.8% (19,768/20,000).

Freezing the embedding removes ~2M parameters from gradient updates, so frozen variants have roughly half the trainable parameters.

Results

Model	Params (scratch/frozen)	Scratch PPL	W2V PPL	GloVe PPL
RNN	4.06M / 2.06M	96.26	93.13*	111.39
LSTM	4.18M / 2.18M	78.00	75.06*	83.12
Transformer	4.28M / 2.28M	53.83*	57.65	60.76

*Table 3. Test perplexity by embedding variant. * = best per model.*

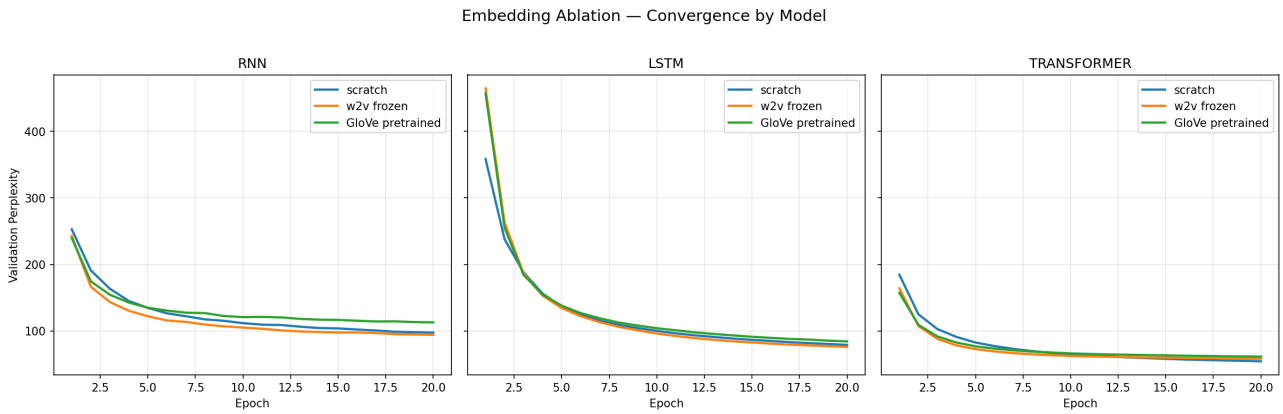

Figure 3. Validation perplexity over 20 epochs for each model-embedding combination.

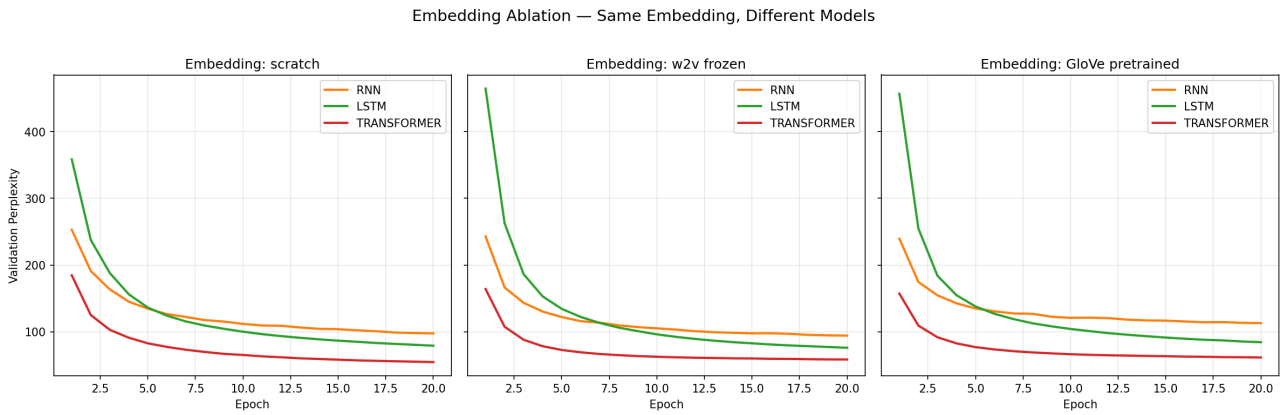

Figure 4. Holding the embedding constant, the Transformer consistently outperforms LSTM and RNN.

Convergence Speed

To assess how quickly each configuration learns, we compare epoch-1 validation perplexity (a proxy for how useful the initial embeddings are) against the final best.

Run	Epoch-1 PPL	Best PPL	Gap
RNN scratch	252.80	97.46	155.34
RNN W2V	242.86	94.18	148.68
RNN GloVe	239.55	112.84	126.71
LSTM scratch	358.39	79.01	279.38
LSTM W2V	464.46	75.96	388.51
LSTM GloVe	456.77	84.28	372.49
TF scratch	184.57	54.42	130.16
TF W2V	164.01	58.37	105.65
TF GloVe	157.18	61.39	95.79

Table 4. Epoch-1 perplexity as a convergence speed indicator.

Analysis

W2V dominates for recurrent models. W2V frozen achieves the best test perplexity for both RNN (93.1 vs 96.3 scratch) and LSTM (75.1 vs 78.0). W2V was self-trained on AG News, so it encodes domain-specific co-occurrence patterns with near-perfect coverage. GloVe, despite a much larger training corpus, encodes general-purpose semantics that are less aligned with news text.

Scratch wins for Transformer. Self-attention reshapes representations at every layer, effectively learning task-optimal embeddings during training. With 2M additional trainable embedding parameters, scratch Transformer has more capacity to

specialize.

LSTM convergence anomaly. LSTM scratch (epoch-1 PPL 358) starts better than frozen variants (W2V 464, GloVe 457) -- the opposite of RNN and Transformer. LSTM gate initialization is calibrated for random-scale inputs; pretrained embeddings with different scale cause suboptimal initial gate behavior. Despite the slow start, W2V LSTM still converges to the best final perplexity.

GloVe is worst everywhere. Domain mismatch (general vs news) and lower coverage (98.8% vs 99.98%) outweigh the benefit of larger pretraining data.

Part III: Downstream Task with Learned Representations

Having trained language models of varying quality, we ask: do better language models produce better features for classification? We extract frozen representations from each trained LM and use them for AG News topic classification.

Feature Extraction and Methods

For each LM, we run a forward pass over every document and extract hidden states at the final layer, mean-pooled across non-padding positions to produce a 100-dimensional feature vector per document.

Method	Description
BoW + LogReg	CountVectorizer + logistic regression (sklearn)
Frozen LM + Linear	Frozen features -> linear layer (100->4)
Frozen LM + MLP	Frozen features -> MLP (100->128->4, ReLU)
Fine-tuned TF + Linear	End-to-end fine-tuning, diff. LR (LM:1e-4, head:1e-3)

Table 5. Downstream classification methods.

We apply frozen probes to all three backbones and fine-tune only the Transformer as the strongest backbone. All classifiers train for 10 epochs with Adam.

Results

Method	Acc	F1	World	Sports	Business	Sci/Tech
BoW + LogReg	0.902	0.902	0.906	0.961	0.865	0.877
RNN frozen+Lin	0.843	0.842	0.848	0.922	0.796	0.803
RNN frozen+MLP	0.858	0.857	0.864	0.935	0.811	0.821
LSTM frozen+Lin	0.873	0.872	0.871	0.945	0.825	0.848
LSTM frozen+MLP	0.884	0.884	0.883	0.947	0.842	0.862
TF frozen+Lin	0.878	0.878	0.884	0.944	0.839	0.846
TF frozen+MLP	0.906	0.906	0.913	0.965	0.867	0.877
TF fine-tuned	0.931	0.930	0.940	0.978	0.901	0.903

Table 6. Downstream results. Per-class columns show F1 scores.

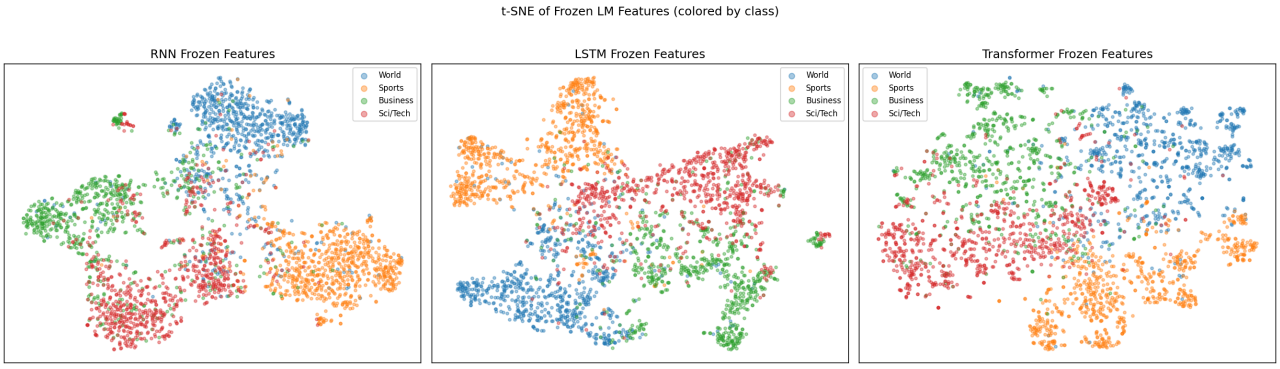

Figure 5. t-SNE of frozen LM features (3,000 test samples). Transformer features show the clearest cluster separation.

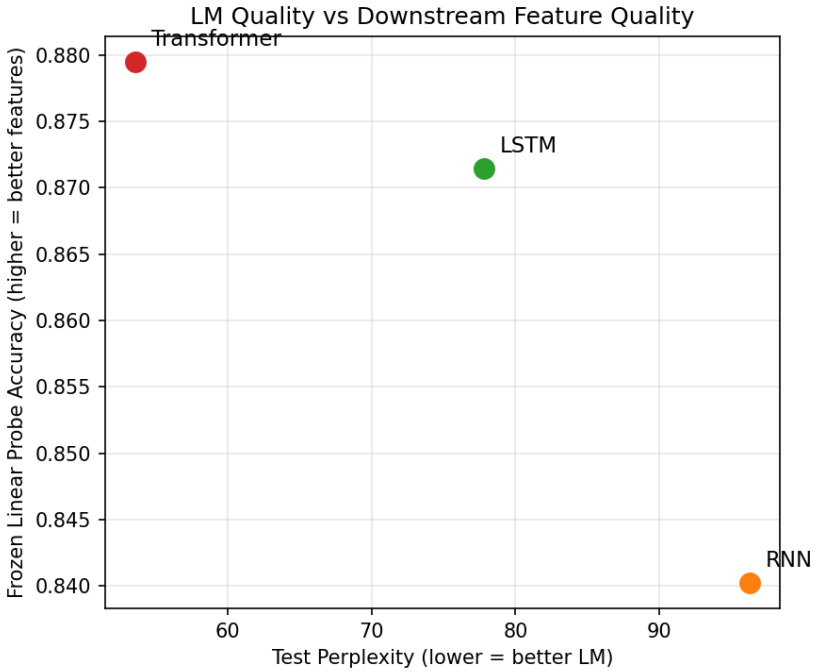

Figure 6. Perplexity vs frozen linear probe accuracy. Better LMs produce better features, with diminishing returns.

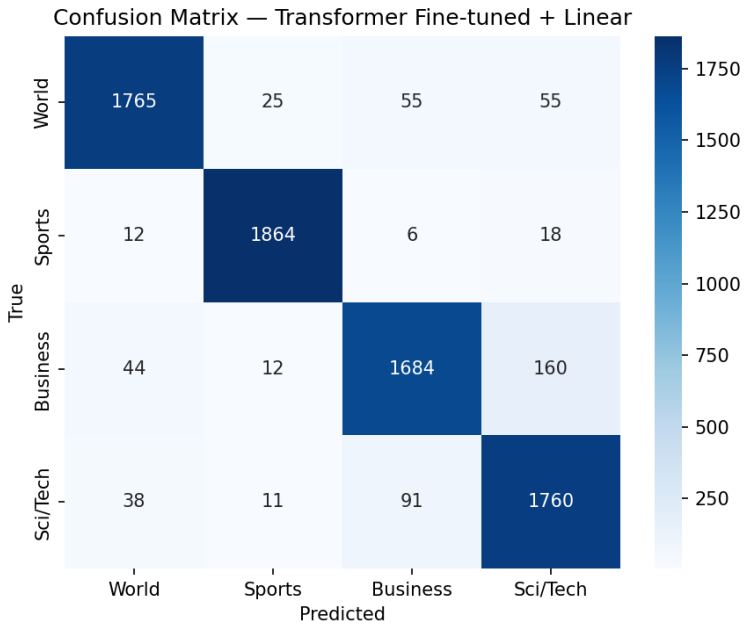

Figure 7. Confusion matrix for the fine-tuned Transformer (best model).

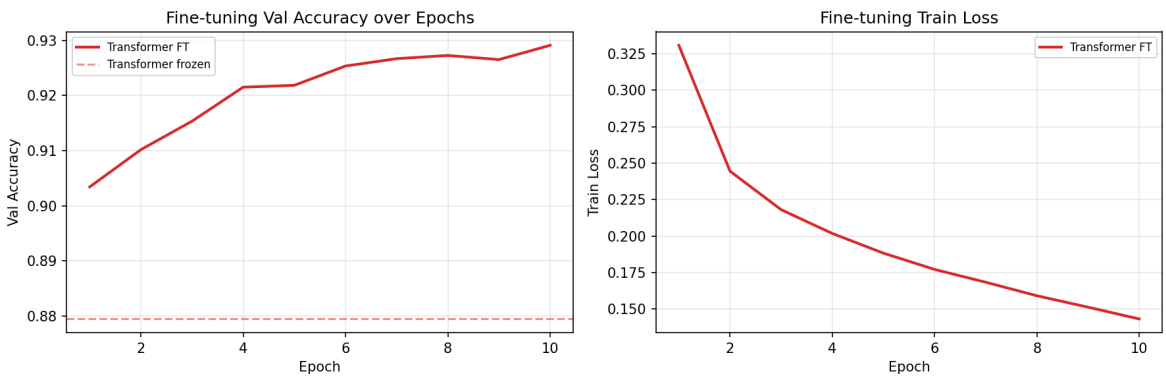

Figure 8. Fine-tuning dynamics. Val accuracy rises from 90.1% to 92.8% with no overfitting.

Analysis

BoW outperforms all frozen linear probes. A simple bag-of-words (90.2%) beats every frozen LM + linear head, including

the Transformer (87.8%). Topic classification depends on which words appear, and BoW captures this directly. Frozen LM features, shaped by next-token prediction, do not explicitly encode class-discriminative information.

Nonlinear heads unlock the Transformer. MLP gains are uneven: RNN +1.5%, LSTM +1.1%, Transformer +2.7%. The larger gain indicates Transformer features encode richer nonlinear structure from self-attention. TF frozen + MLP (90.6%) is the only frozen config surpassing BoW.

PPL correlates with feature quality, with diminishing returns. RNN (PPL 96.3, Acc 84.3%) -> LSTM (PPL 77.8, Acc 87.3%) -> Transformer (PPL 53.6, Acc 87.8%). The 31% PPL drop LSTM->TF yields only +0.6% linear probe accuracy, as further perplexity gains capture patterns that do not help coarse topic classification.

Fine-tuning gives the largest gain. The fine-tuned Transformer (93.1%) gains +5.2% over frozen + linear and +2.5% over frozen + MLP. Fine-tuning reshapes representations from "what word comes next" to "what topic is this." Val accuracy climbs from 90.1% to 92.8% over 10 epochs with no overfitting.

Per-class patterns. Sports is easiest (F1 0.92-0.98); Business is hardest (F1 0.80-0.90). Fine-tuning helps weak classes most: Business F1 improves from 0.84 to 0.90 (+7% absolute).

Connection to LM architecture. More expressive architectures produce richer representations, but that richness is only accessible through nonlinear probes or fine-tuning. Recurrent models compress through a fixed-size hidden state, yielding more linearly separable features. The Transformer distributes information across a higher-dimensional attention space that requires more downstream capacity to exploit -- but achieves the best results when given that capacity.

Experimental Controls

- Parameter fairness: All neural LMs have 4.06-4.28M parameters. The shared embedding (20K x 100 = 2M) and output projection (100 x 20K = 2M) dominate; differences come only from model cores.
- Same pipeline: Same tokenizer, vocabulary, splits, and max sequence length (128).
- No data leakage: W2V trained only on the training split. Zero split overlap verified.
- Sanity checks: All models overfit one batch (>50% loss drop); all parameters receive gradients; Transformer causal mask verified.
- Reproducibility: Seed 42. Hyperparameters centralized in configs/default.yaml.